Ethical Reasoning Mapper (ERM)

Coding Manual for Open-Ended Responses to Ethical Dilemmas

TECHNICAL MANUAL

Version 1.1

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1. Purpose

The Ethical Reasoning Mapper (ERM) specifies a standardized coding framework for the empirical analysis of ethical reasoning expressed in open-ended responses to ethical dilemmas.

The ERM provides a common coding specification for identifying, classifying, and documenting the explicit components of ethical reasoning through a predefined set of operational variables. Its primary purpose is to support transparent, reproducible, and internally consistent analyses across qualitative research and educational settings.

The framework is organized into three complementary analytical dimensions:

- Ethical Reasoning Components, identifying the explicit components through which ethical reasoning is constructed, including ethical positions, factual and normative premises, conclusions, counterarguments, moral conflicts, deliberative weighing, and proposed resolutions.
- Morally Salient Considerations, identifying the values, principles, interests, rights, relationships, and other considerations that participants explicitly recognize as morally relevant to the ethical dilemma.
- Sources of Justification, documenting the grounds upon which ethical judgments are supported, including empirical evidence, professional knowledge, philosophical concepts, legal reasoning, personal experience, cultural traditions, religious beliefs, and other identifiable sources of normative authority.

The ERM is designed for the analysis of written or transcribed responses generated in educational, research, and professional contexts in which participants are asked to reason about ethical or moral dilemmas.

The ERM is a descriptive analytical instrument. It does not evaluate the moral correctness of a response, determine the adequacy of a philosophical position, assess moral competence, or measure educational achievement. Rather, it provides a standardized methodology for reconstructing the explicit architecture of ethical reasoning as expressed in discourse.

2. Scope

The Ethical Reasoning Mapper (ERM) is intended for the analysis of ethical reasoning elicited through open-ended responses to ethical or moral dilemmas.

The framework is applicable to qualitative data generated through written responses, oral interviews, reflective essays, classroom discussions, examinations, case analyses, focus groups, and other elicitation methods in which participants are invited to formulate and justify ethical judgments.

The ERM is designed for educational, research, and professional settings in which ethical reasoning constitutes an object of empirical inquiry. Typical applications include bioethics, medicine, public health, philosophy, law, public policy, engineering, artificial intelligence, environmental ethics, and other disciplines involving ethical decision-making.

Although the ERM was originally developed using responses to dilemmas concerning human–animal relations, its coding specification is intended for any context in which participants engage in ethical reasoning about contested normative questions. The framework is independent of the substantive topic under discussion and may be applied across different populations, educational levels, and disciplinary contexts.

The ERM is intended for qualitative and mixed-methods research based on the systematic coding of ethical reasoning. It is not designed for automated text classification, psychometric assessment, evaluation of moral competence, or the determination of normative correctness.

3. Conceptual Framework

Ethical reasoning has been conceptualized from multiple theoretical traditions, including moral development, ethical deliberation, argumentation theory, professional judgment, moral psychology, and value theory (Beauchamp & Childress, 2019; Gracia, 2019; Greene, 2013; Kohlberg, 1984; Rest et al., 1999). Although these traditions differ in their normative assumptions and analytical focus, they converge in recognizing that ethical reasoning involves more than the expression of a moral opinion. Ethical reasoning entails the articulation of reasons, the identification of morally relevant considerations, and the justification of ethical judgments.

The Ethical Reasoning Mapper (ERM) operationalizes ethical reasoning through three complementary analytical dimensions. Each dimension captures a distinct component of ethical reasoning and is grounded in a different theoretical tradition. Together, these dimensions provide a comprehensive framework for reconstructing the explicit architecture of ethical reasoning expressed in response to ethical dilemmas.

Ethical Reasoning Components

The first dimension identifies the explicit components through which ethical reasoning is constructed. It is informed by theories of argumentation and informal logic, particularly Toulmin's model of practical argumentation (Toulmin, 2003), while extending beyond formal argument structure to accommodate the characteristics of ethical discourse.

Rather than evaluating the validity of an argument, this dimension identifies whether a response explicitly contains elements such as ethical positions, factual premises, normative premises, conclusions, counterarguments, moral conflicts, deliberative weighing, and proposed resolutions. The objective is to reconstruct the observable architecture of ethical reasoning as it appears in discourse.

Morally Salient Considerations

The second dimension identifies the considerations that participants explicitly recognize as morally relevant when evaluating an ethical dilemma. This dimension draws upon value theory and contemporary approaches to practical ethics, which recognize that ethical judgment depends upon the identification and prioritization of morally significant values rather than the application of a single normative rule (Gracia, 2019; Nussbaum, 2011; Sen, 2009).

Instead of assigning responses to predefined ethical schools, the ERM documents the values, principles, interests, rights, vulnerabilities, relationships, and other considerations that participants mobilize during ethical reasoning.

Sources of Justification

The third dimension documents the explicit sources upon which ethical judgments are justified. Ethical reasoning may be supported by empirical evidence, professional expertise, philosophical concepts, legal norms, personal experience, cultural traditions, religious beliefs, or other identifiable forms of justification. This dimension parallels the function of 'backing' in Toulmin's model of argumentation — the body of support invoked to establish the acceptability of a warrant (Toulmin, 2003).

This dimension is descriptive rather than normative. Its purpose is not to evaluate the adequacy of a justification, but to identify the sources participants invoke to support their ethical judgments and to characterize patterns of justification across individuals or contexts.

Although analytically independent, the three dimensions are complementary. Ethical reasoning is understood as the integration of reasoning components, morally salient considerations, and sources of justification into a coherent ethical judgment. The ERM reconstructs these dimensions independently to facilitate transparent, reproducible, and comparable analyses of ethical reasoning across educational and research settings.

4. Unit of Analysis

The primary unit of analysis in the Ethical Reasoning Mapper (ERM) is the individual response to a single ethical or moral dilemma.

A response is defined as the complete verbal or written discourse produced by a participant in relation to one prompt or case. Each response constitutes an independent analytical unit and is coded separately.

The ERM does not code isolated words, sentences, or paragraphs as independent units. Coding is performed at the response level to preserve the coherence of ethical reasoning and to allow the identification of relationships among reasoning components, morally salient considerations, and sources of justification.

Within a response, individual statements may be examined to identify specific coding variables. However, variables are assigned only after considering the response as an integrated whole. Components distributed across multiple sentences are interpreted jointly whenever they contribute to the same line of ethical reasoning.

When a participant provides multiple independent answers to different ethical dilemmas, each answer constitutes a separate unit of analysis. Likewise, repeated responses from the same participant (e.g., pretest and posttest) are treated as independent analytical units while remaining linked through participant identifiers for longitudinal analyses.

The ERM assumes that ethical reasoning may be explicit, incomplete, or internally inconsistent. Consequently, the objective of coding is not to reconstruct an ideal argument but to identify the explicit components of ethical reasoning present within each response.

> **Coding Rule 1**
>
> ***Code what is explicitly expressed, not what can be inferred. Variables are assigned exclusively on the basis of information explicitly articulated by the participant. Coders should not infer missing premises, conclusions, values, or sources of justification unless they are directly expressed in the response.***

5. Data Structure

The Ethical Reasoning Mapper (ERM) is based on a rectangular data structure in which each row represents a single analytical unit and each column represents a predefined variable. This rectangular, category-based structure follows standard practice in quantitative content analysis (Krippendorff, 2018). The ERM is software-independent. The data structure may be implemented in spreadsheets, relational databases, statistical software, qualitative data analysis software, or reproducible analytical workflows developed in programming languages such as R or Python.

The recommended analytical unit is one participant's response to one ethical dilemma. Consequently, each observation corresponds to a unique combination of participant, assessment occasion, dilemma, and response.

The minimum required variables are:

Variable	Description
response_id	Unique identifier for the response.
participant_id	Unique identifier for the participant.
timepoint	Assessment occasion (e.g., Pretest, Posttest, Follow-up).
dilemma_id	Identifier of the ethical dilemma.
response_text	Original verbatim response.

The ERM variables are recorded in separate columns following the metadata. Each analytical variable occupies an independent column and is coded according to the specifications provided in Section 8.

The framework is designed primarily for binary coding (0 = absent; 1 = present). Future versions may incorporate ordinal or nominal variables where justified by empirical validation.

A simplified data structure is illustrated below.

response_id	participant_id	timepoint	dilemma_id	response_text	ERC_01	ERC_02	MSC_01	SJ_01
R001	P001	Pre	D01	…	1	0	1	0
R002	P001	Post	D01	…	1	1	1	1
R003	P002	Pre	D01	…	0	1	0	1

Variables are grouped according to the three analytical dimensions:

- ERC — Ethical Reasoning Components
- MSC — Morally Salient Considerations
- SJ — Sources of Justification

Each variable is coded independently. Multiple variables within the same dimension may be coded as present in a single response. The absence of one component does not preclude the presence of others.

6. Coding Workflow

The Ethical Reasoning Mapper (ERM) follows a standardized sequential workflow designed to ensure consistent and reproducible coding across coders, datasets, and analytical contexts.

The recommended workflow comprises seven stages.

Step 1. Data Preparation

Prepare the dataset according to the ERM data structure described in Section 5. Each analytical unit must correspond to a single participant's response to one ethical dilemma and contain a unique identifier.

Step 2. Familiarization

Read the complete response before assigning any coding variables.

Responses should be interpreted as integrated units of ethical reasoning. Coding individual sentences without considering the complete response is discouraged.

Step 3. Identification of Ethical Reasoning Components

Identify the explicit components of ethical reasoning present in the response according to the specifications of the Ethical Reasoning Components dimension.

Multiple components may be identified within the same response.

Step 4. Identification of Morally Salient Considerations

Identify all values, principles, interests, rights, relationships, vulnerabilities, and other considerations explicitly recognized as morally relevant by the participant.

Each consideration is coded independently.

Step 5. Identification of Sources of Justification

Identify every explicit source used to justify the ethical judgment.

Multiple sources of justification may coexist within the same response.

Step 6. Quality Review

Review all assigned variables to verify internal consistency and compliance with the coding specifications.

Whenever multiple coders participate, discrepancies should be resolved according to the procedures described in Section 10.

Step 7. Data Export

Export the coded dataset in a structured format suitable for statistical analysis, visualization, or qualitative interpretation.

Workflow Summary

Data Preparation → Familiarization → Ethical Reasoning Components → Morally Salient Considerations → Sources of Justification → Quality Review → Data Export

> **Coding Principle 1**
>
> ***Responses are coded sequentially, not simultaneously. Coders should complete each analytical dimension before proceeding to the next. Coding multiple dimensions simultaneously increases the risk of interpretative bias and inconsistent variable assignment.***

7. Coding Dimensions

The Ethical Reasoning Mapper (ERM) reconstructs ethical reasoning through three complementary analytical dimensions. Each dimension addresses a distinct analytical question and is coded independently using its own operational variables.

The dimensions are sequentially organized to reflect the analytical logic of the framework. Coders first identify the explicit components through which ethical reasoning is constructed, then identify the morally salient considerations mobilized within that reasoning, and finally document the sources upon which those considerations are justified.

Although analytically distinct, the three dimensions collectively characterize the explicit architecture of ethical reasoning expressed in response to an ethical dilemma.

Ethical Reasoning Components

This dimension identifies the explicit components through which ethical reasoning is constructed. Rather than evaluating argumentative quality or logical validity, the ERM documents whether specific reasoning components are explicitly articulated within a response. These components include ethical positions, factual premises, normative premises, conclusions, counterarguments, moral conflicts, deliberative weighing, and proposed resolutions.

The objective of this dimension is to reconstruct the explicit structure of ethical reasoning without inferring unstated premises or conclusions.

Morally Salient Considerations

This dimension identifies the considerations that participants explicitly recognize as morally relevant to the ethical dilemma.

Morally salient considerations include values, principles, rights, interests, responsibilities, vulnerabilities, relationships, capabilities, and other ethical considerations that participants invoke when constructing their reasoning.

The ERM does not classify responses according to predefined ethical theories. Instead, it documents the considerations that participants themselves explicitly mobilize during ethical reasoning.

Sources of Justification

This dimension identifies the explicit sources upon which ethical judgments are justified.

Sources of justification may include empirical evidence, scientific knowledge, professional expertise, philosophical concepts, legal norms, personal experience, cultural traditions, religious beliefs, analogical reasoning, or other explicitly identifiable foundations.

This dimension is descriptive. It documents how participants justify their ethical judgments without evaluating the epistemic strength or normative adequacy of those justifications.

8. Variable Specifications

The ERM comprises 27 operational variables organized into three analytical dimensions: 8 Ethical Reasoning Components (ERC), 10 Morally Salient Considerations (MSC), and 9 Sources of Justification (SJ). Each variable is independently coded according to the specifications presented below.

Unless otherwise specified:

- 0 = Absent
- 1 = Present

Variables are assigned exclusively on the basis of explicitly expressed information.

Ethical Reasoning Components (ERC)

ID	Variable	Operational Definition	Coding Criteria	Notes
ERC01	Ethical Position	An explicit normative stance regarding the ethical dilemma.	Code 1 when the participant explicitly states what ought or ought not to be done or evaluates an action, institution, or practice in moral terms.	Descriptive statements alone do not qualify. See Decision Tree 1 (Appendix B) for the boundary with ERC04.
ERC02	Factual Premise	An empirical proposition used to support ethical reasoning.	Code 1 when factual information functions as a premise for the ethical judgment.	Scientific accuracy is not evaluated. See Decision Tree 4 (Appendix B) for the boundary with ERC03.
ERC03	Normative Premise	An explicit moral principle, value, duty, or norm supporting the reasoning.	Code 1 when a normative proposition is explicitly articulated.	Implicit values should not be inferred. See Decision Tree 4 (Appendix B) for the boundary with ERC02.
ERC04	Conclusion	An explicit ethical conclusion derived from the reasoning.	Code 1 when the participant reaches a normative conclusion.	May coincide with the ethical position. See Decision Tree 1 (Appendix B).
ERC05	Counterargument	Recognition of an alternative position or objection.	Code 1 when another perspective is explicitly acknowledged.	Agreement is not required.
ERC06	Moral Conflict	Explicit recognition of competing moral considerations.	Code 1 when two or more moral considerations are presented as being in tension.	Mere listing of values is insufficient.
ERC07	Deliberative Weighing	Explicit comparison or balancing of competing considerations before reaching a conclusion.	Code 1 when alternatives are compared or weighed.	Requires more than acknowledging a conflict.
ERC08	Resolution	Proposal of a justified course of action.	Code 1 when a practical recommendation is explicitly stated.	Recommendations without ethical justification remain coded independently.

Morally Salient Considerations (MSC)

ID	Variable	Operational Definition	Include	Exclude	Notes
MSC01	Welfare / Well-being	Explicit consideration of physical or psychological well-being, suffering, harm, or benefit.	Welfare, pain, suffering, health, quality of life, benefit, harm.	Purely factual descriptions without moral relevance.	Applies to humans, animals, or other morally relevant entities. See Decision Tree 2 (Appendix B) for the boundary with MSC09.
MSC02	Rights	Explicit recognition of rights, entitlements, or moral claims.	Human rights, animal rights, legal or moral rights.	General references to laws without invoking rights.	Rights may be legal or moral. See Decision Tree 5 (Appendix B) for the boundary with MSC04.
MSC03	Justice / Fairness	Explicit concern for fairness, equality, equity, or distributive justice.	Equal treatment, discrimination, fairness, resource allocation.	Personal preferences.	Includes distributive, procedural, and social justice.
MSC04	Autonomy	Explicit recognition of agency, freedom, self-determination, or informed choice.	Consent, freedom, decision-making capacity, self-determination.	Mere descriptions of behavior.	Includes individual and collective autonomy when explicitly stated. See Decision Tree 5 (Appendix B) for the boundary with MSC02.
MSC05	Responsibility / Duty	Explicit appeal to obligations, duties, accountability, or professional responsibilities.	Moral duty, professional obligations, responsibility toward others.	Simple descriptions of actions without normative obligation.	Includes personal and institutional responsibilities. See Decision Tree 6 (Appendix B) for the boundary with SJ04.
MSC06	Vulnerability	Explicit recognition of fragility, dependency, marginalization, or the need for protection.	Vulnerable populations, dependency, inability to protect oneself.	General descriptions of disadvantage without ethical relevance.	Includes structural and situational vulnerability.
MSC07	Relationships / Care	Explicit emphasis on interpersonal relationships, empathy, care, trust, or relational responsibilities.	Caring, compassion, family relationships, professional relationships.	Neutral descriptions of social interaction.	Includes ethics of care and relational responsibilities.
MSC08	Intrinsic Value	Explicit recognition that an individual, organism, or entity possesses value independent of its utility.	Inherent dignity, intrinsic worth, value in itself.	Instrumental usefulness alone.	Particularly relevant in environmental and animal ethics.

ID	Variable	Operational Definition	Include	Exclude	Notes
MSC09	Capabilities / Flourishing	Explicit concern for the ability to develop, function, or flourish.	Human flourishing, capabilities, opportunities to live well.	General references to happiness without normative context.	Consistent with capability-based approaches. See Decision Tree 2 (Appendix B) for the boundary with MSC01.
MSC10	Social or Cultural Context	Explicit recognition that social, cultural, historical, political, or institutional conditions are morally relevant.	Social inequality, culture, institutions, historical context, structural factors.	Context provided only as background information.	Code only when context is explicitly incorporated into the ethical reasoning.

Sources of Justification (SJ)

ID	Variable	Operational Definition	Coding Criteria	Notes
SJ01	Empirical / Scientific Evidence	Explicit reliance on empirical data, research findings, or scientific knowledge to support the ethical judgment.	Code 1 when the participant cites data, studies, or scientific facts as grounds for the judgment.	Accuracy of the evidence is not evaluated. See Decision Tree 3 (Appendix B) for the boundary with SJ08.
SJ02	Professional / Clinical Expertise	Explicit appeal to professional training, clinical experience, or disciplinary expertise as justification.	Code 1 when professional knowledge or practice is invoked as the basis for the judgment.	Includes references to professional guidelines or standards of practice.
SJ03	Philosophical / Theoretical Concepts	Explicit use of philosophical theories, ethical frameworks, or theoretical concepts to justify the judgment.	Code 1 when a named theory, framework, or philosophical concept is explicitly invoked.	The concept must be named or clearly identifiable, not merely implied.
SJ04	Legal / Regulatory Norms	Explicit appeal to laws, regulations, or formal institutional norms.	Code 1 when a legal or regulatory reference is used to justify the judgment.	Includes references to policy or institutional rules. See Decision Tree 6 (Appendix B) for the boundary with MSC05.

ID	Variable	Operational Definition	Coding Criteria	Notes
SJ05	Personal Experience	Explicit reference to the participant's own lived experience as grounds for the judgment.	Code 1 when the participant explicitly draws on a personal experience or anecdote.	Excludes general statements without a specific personal referent.
SJ06	Cultural / Traditional Norms	Explicit appeal to cultural practices, customs, or tradition.	Code 1 when the judgment is grounded in convention or 'how things are usually done.'	Distinguish from MSC10, which codes cultural context as a morally salient consideration rather than a justificatory
SJ07	Religious / Spiritual Beliefs	Explicit appeal to religious doctrine, spiritual belief, or faith-based reasoning.	Code 1 when a religious or spiritual source is explicitly named as grounds for the judgment.	Requires an explicit religious or spiritual referent, not general moral language.
SJ08	Analogical Reasoning	Explicit use of an analogy or comparison to a different case, situation, or entity to justify the judgment.	Code 1 when the participant compares the dilemma to another case, situation, or precedent to support their reasoning.	The analogy must be explicit, not merely implied by juxtaposition. See Decision Tree 3 (Appendix B) for the boundary with SJ01.
SJ09	Other Explicit Source	Any other explicitly identifiable source of justification not captured by SJ01–SJ08.	Code 1 when a distinct, explicit source of justification is present but does not fit the preceding categories.	Coders should annotate the specific source identified, for later review and potential codebook refinement.

9. Coding Rules

The following rules apply across all three analytical dimensions of the ERM and govern how coders assign, combine, and document variables. They formalize and extend the principle introduced in Section 4 (Coding Rule 1).

- Rule 1. Code what is explicitly expressed, not what can be inferred. Coders should not infer missing premises, conclusions, values, or sources of justification unless they are directly expressed in the response.

- Rule 2. Variables are not mutually exclusive across dimensions. A single response may simultaneously receive codes in the ERC, MSC, and SJ dimensions, since a response typically articulates a component of reasoning, invokes a morally salient consideration, and grounds it in a source of justification at the same time.
- Rule 3. Within a dimension, more than one variable may be coded as present. For example, a response may simultaneously invoke MSC01 (Welfare) and MSC03 (Justice); the presence of one consideration does not preclude another.
- Rule 4. Coding is binary at the level of the response, not a frequency count within the response. A variable is coded present only once per response, regardless of how many times the corresponding element recurs within that response.
- Rule 5. The presence of a component is coded independently of the participant's stance toward it. For example, explicit rejection of a counterargument still constitutes engagement with a counterargument (ERC05 = 1); coders code whether the component is present, not whether the participant agrees with it.
- Rule 6. When a response addresses multiple sub-dilemmas within a single prompt, the response is coded as a whole, consistent with the unit of analysis defined in Section 4, unless the ERM has been formally adapted to a finer unit of analysis.
- Rule 7. Boundary or ambiguous cases should first be checked against the decision trees in Appendix B. If no applicable decision tree resolves the ambiguity, the case should be logged and discussed as part of the quality control procedure described in Section 10.
- Rule 8. Coding decisions and their rationale should be documented for borderline cases, to support auditability, coder training, and future codebook refinement.
- Rule 9. Instances coded as SJ09 (Other Explicit Source) should be reviewed collectively at each reliability checkpoint (Section 10). If a recurring, distinguishable source accounts for at least 5% of coded responses within a checkpoint interval, the coding team should consider formalizing it as a new SJ variable in a subsequent codebook revision, following the version-control procedure documented in Appendix F.

10. Quality Control

The ERM recommends the following quality control procedures to ensure coding reliability and analytical transparency.

Coder training

Coders should be trained on the complete codebook (Section 8 and Appendix A) prior to independent coding, including review of the worked examples (Section 11) and the annotated responses (Appendix C).

Pilot coding

Before full-scale coding, a pilot subsample should be independently coded by at least two coders, to identify ambiguities in the codebook and refine variable definitions if necessary. The recommended pilot size is 10–15% of the corpus, subject to a minimum of 30 coded responses (or the full corpus, whichever is smaller), consistent with sample-size guidance for kappa-based agreement studies (Bujang & Baharum, 2017) and established practice in intercoder-reliability research (O’Connor & Joffe, 2020).

Inter-rater reliability

Following the pilot phase, inter-rater reliability should be assessed for each variable independently. For two coders, use the unweighted Cohen's kappa appropriate for nominal binary data (Cohen, 1960; e.g., irr::kappa2(data, weight = "unweighted") in R). For more than two coders, or where ratings are missing, use Krippendorff's alpha computed at the nominal level of measurement (Krippendorff, 2018; Hayes & Krippendorff, 2007; e.g., irr::kripp.alpha(data, method = "nominal") in R). A minimum threshold of $\kappa \geq .70$ (or $\alpha \geq .70$) per variable is recommended before proceeding to full-scale coding, consistent with the conventional 'substantial agreement' band (Landis & Koch, 1977); this benchmark is a widely used convention rather than a fixed statistical criterion, and coders should interpret it cautiously for variables with low base-rate prevalence, where kappa is known to become unstable even at high observed agreement. Variables falling below this threshold should be revised and re-piloted.

Discrepancy resolution

Disagreements identified during reliability assessment should be resolved through consensus discussion among coders. Where consensus cannot be reached, a third coder or the lead researcher should adjudicate, and the resolution should be documented.

Ongoing monitoring

For large corpora, periodic reliability checks (e.g., every 20% of the corpus coded) are recommended to detect coder drift over time. If a checkpoint reliability estimate for any variable falls below the $\kappa \geq .70$ (or $\alpha \geq .70$) threshold, coding of that variable should pause while the corresponding codebook entry is reviewed and clarified, followed by a new pilot round before full-scale coding resumes.

Audit trail

All coding decisions, codebook revisions, and discrepancy resolutions should be logged to preserve the traceability of the analytical process, consistent with standards of transparency in qualitative research (Lincoln & Guba, 1985).

Validity Considerations

In addition to reliability, the content validity of the ERM codebook should be established before deployment. The recommended procedure is expert panel review: at least three subject-matter experts (e.g., in bioethics, moral psychology, or the relevant applied domain) should independently assess whether each of the 27 variables (a) reflects a conceptually distinct, non-redundant construct, and (b) is exhaustive with respect to the theoretical dimension to which it belongs, given the intended context of application. Disagreements should be resolved through the same discrepancy-resolution procedure used for coding, and the codebook revised accordingly before piloting. Because the ERM is a descriptive coding instrument rather than a psychometric scale, criterion validity against an external gold standard is not applicable; content and face validity, together with inter-rater reliability, constitute the primary evidentiary basis for its use (cf. Krippendorff, 2018).

Validation Status of This Version

As of Version 1.1, the ERM has not yet undergone the empirical validation procedures it recommends: no expert panel review, pilot coding, or inter-rater reliability assessment has been conducted on this codebook. The framework is proposed pending such validation. Researchers who adopt the ERM are encouraged to conduct and report pilot reliability results (Section 10) and to share them with the authors to inform future codebook revisions (Appendix F).

11. Worked Examples

This section illustrates the coding process using an example response. Appendix C provides additional annotated responses for training purposes.

Consider the following illustrative response (constructed for demonstration purposes, not drawn from a specific study) to a dilemma concerning the use of animals in scientific research:

"I think using animals in research is only acceptable when there is no alternative and the suffering is minimized. Animals can feel pain, so causing them harm needs a strong justification — similar to how we would justify a difficult medical procedure on a person who cannot consent, if it saves many lives. But I do not think that justifies cosmetic testing, because the benefit does not outweigh the harm."

Applying the ERM:

- ERC01 (Ethical Position) = 1 — the participant states what is and is not acceptable.

- ERC03 (Normative Premise) = 1 — "causing harm needs a strong justification" functions as an explicit normative principle.
- ERC06 (Moral Conflict) = 1 — benefit and harm are explicitly framed as competing considerations.
- ERC07 (Deliberative Weighing) = 1 — the benefit is explicitly weighed against the harm ("the benefit does not outweigh the harm").
- ERC08 (Resolution) = 1 — a practical conclusion is proposed (acceptable for research that saves lives, not for cosmetic testing).
- MSC01 (Welfare / Well-being) = 1 — explicit reference to pain and suffering.
- SJ08 (Analogical Reasoning) = 1 — explicit comparison to a medical procedure on a non-consenting person.

This example illustrates how a single response can receive multiple codes across dimensions, consistent with Rules 2 and 3 (Section 9).

12. Variable Dictionary

The following table provides a compact index of all 27 ERM variables, for quick reference during coding. Full operational definitions and coding criteria are provided in Section 8 and consolidated in Appendix A.

ID	Short Label	Dimension
ERC01	Ethical Position	**Ethical Reasoning Components**
ERC02	Factual Premise	**Ethical Reasoning Components**
ERC03	Normative Premise	**Ethical Reasoning Components**
ERC04	Conclusion	**Ethical Reasoning Components**
ERC05	Counterargument	**Ethical Reasoning Components**
ERC06	Moral Conflict	**Ethical Reasoning Components**
ERC07	Deliberative Weighing	**Ethical Reasoning Components**
ERC08	Resolution	**Ethical Reasoning Components**
MSC01	Welfare / Well-being	**Morally Salient Considerations**
MSC02	Rights	**Morally Salient Considerations**
MSC03	Justice / Fairness	**Morally Salient Considerations**
MSC04	Autonomy	**Morally Salient Considerations**
MSC05	Responsibility / Duty	**Morally Salient Considerations**
MSC06	Vulnerability	**Morally Salient Considerations**
MSC07	Relationships / Care	**Morally Salient Considerations**
MSC08	Intrinsic Value	**Morally Salient Considerations**

ID	Short Label	Dimension
MSC09	Capabilities / Flourishing	**Morally Salient Considerations**
MSC10	Social or Cultural Context	**Morally Salient Considerations**
SJ01	Empirical / Scientific Evidence	**Sources of Justification**
SJ02	Professional / Clinical Expertise	**Sources of Justification**
SJ03	Philosophical / Theoretical Concepts	**Sources of Justification**
SJ04	Legal / Regulatory Norms	**Sources of Justification**
SJ05	Personal Experience	**Sources of Justification**
SJ06	Cultural / Traditional Norms	**Sources of Justification**
SJ07	Religious / Spiritual Beliefs	**Sources of Justification**
SJ08	Analogical Reasoning	**Sources of Justification**
SJ09	Other Explicit Source	**Sources of Justification**

13. Data Outputs

The ERM supports several standard analytical outputs once responses have been coded:

- Variable-level frequencies. Proportion of responses in which each variable is coded present, by group, timepoint, or dilemma, reported with exact binomial or Wilson confidence intervals given the binary nature of the underlying codes.
- Dimension indices. For each response, a summary count of variables coded present within a dimension (e.g., an ERC index ranging from 0 to 8), providing a simple measure of argumentative complexity. Because these are bounded counts rather than continuous, interval-scaled measures, report medians and interquartile ranges alongside means, and treat the indices as ordinal-like when selecting statistical tests.
- Co-occurrence patterns. Cross-tabulations identifying which variables tend to co-occur within the same response (e.g., MSC01 with SJ01), reported as pairwise co-occurrence proportions or phi coefficients. Given the large number of pairwise comparisons across 27 variables (351 pairs), any significance tests of association (e.g., Fisher's exact test) should be corrected for multiple comparisons using a false discovery rate procedure (Benjamini & Hochberg, 1995) rather than an uncorrected threshold.
- Longitudinal comparisons. Pre–post (or multi-timepoint) comparisons of variable frequencies and dimension indices at the individual or group level, supporting trajectory analyses. For a single binary variable compared across two timepoints within the same participants, McNemar's test for correlated proportions is appropriate (McNemar, 1947). For dimension indices compared across two timepoints, the Wilcoxon signed-rank test is recommended over a paired t-test given the discrete, bounded, likely non-normal distribution of index scores (Wilcoxon, 1945). For more than two timepoints, or where participants contribute responses to multiple dilemmas, a mixed-effects model with participant as a random effect is recommended to account for the non-independence of repeated responses.
- Participant profiles. Individual-level summaries across all coded responses, useful for case-based or trajectory analysis. Because profiles may aggregate unequal numbers of responses per participant, group-

level analyses based on participant profiles should account for this clustering (e.g., via mixed-effects models or cluster-robust standard errors) rather than treating each response as an independent observation.

These outputs are descriptive and are intended to support, not replace, qualitative interpretation of the coded corpus.

14. R Workflow

Appendix E provides a complete, reproducible R script implementing the workflow summarized below.

- 1. Import the coded dataset (Section 5 structure) into R using readr::read_csv() or readxl::read_excel().
- 2. Validate that all ERC, MSC, and SJ columns are binary (0/1) using dplyr::summarise(across(...)).
- 3. Compute dimension indices by summing variables within each dimension per response.
- 4. Compute variable-level frequencies by group, timepoint, or dilemma using dplyr::group_by() and summarise(), with confidence intervals via, e.g., binom::binom.confint().
- 5. Compute inter-rater reliability statistics using the irr package: irr::kappa2(data, weight = "unweighted") for two coders (Cohen, 1960), or irr::kripp.alpha(data, method = "nominal") for more than two coders or missing ratings (Krippendorff, 2018).
- 6. Test longitudinal change using stats::mcnemar.test() for binary variables (McNemar, 1947) and wilcox.test(..., paired = TRUE) for dimension indices (Wilcoxon, 1945); adjust pairwise co-occurrence tests with stats::p.adjust(method = "BH") (Benjamini & Hochberg, 1995).
- 7. Visualize frequencies and dimension indices using ggplot2.
- 8. Export summary tables and figures for reporting.

15. File Structure

A recommended project structure for ERM-based coding projects:

```
project/

├── data/

│   ├── raw/              # Original response data

│   └── coded/            # ERM-coded dataset(s)

├── codebook/

│   └── ERM_codebook_vX.X.docx
```

```
├── scripts/
│   └── erm_analysis.R
├── outputs/
│   ├── figures/
│   └── tables/
└── README.md
```

This structure describes a typical ERM-based coding project for a new study. The reference materials for this manual itself — the example dataset, the variable dictionary, and the analysis script — are distributed separately as Companion Materials (see below).

16. Version History

Version	Date	Description
1.0	August 2026	Initial complete release, prepared for open deposit. See Appendix F for a detailed changelog.
1.1	August 2026	Strengthened methodological, conceptual, and statistical precision throughout (pilot-size justification, kappa/alpha specification, validity considerations, three new decision trees, statistical tests for Section 13 outputs); added Alma Delia Torres-Rivera as co-author; added a Companion Materials section pointing to the versioned GitHub repository. See Appendix F for a detailed changelog.

Companion Materials

The complete companion materials for Ethical Reasoning Mapper (ERM) version 1.1 are available in the versioned GitHub release:

Repository: https://github.com/steresadiaz/Ethical-Reasoning-Mapper

Version 1.1 release: https://github.com/steresadiaz/Ethical-Reasoning-Mapper/releases/tag/v1.1

The repository includes:

- the complete variable dictionary in machine-readable format;
- the example dataset;
- the production-ready R analysis script;
- data-validation and reliability-analysis utilities;
- the project README, license information, and version history.

For reproducibility, users should cite and download the v1.1 release rather than the repository's default branch. The materials associated with later releases may differ from those described in this manual.

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Appendix A. Complete Codebook

This appendix consolidates all 27 ERM variables into a single reference table, intended for use as a quick-lookup coding card. Full coding criteria, inclusion/exclusion notes, and worked examples are provided in Section 8, Section 11, and Appendix C.

ID	Dimension	Variable	Operational Definition
ERC01	**ERC**	Ethical Position	An explicit normative stance regarding the ethical dilemma.
ERC02	**ERC**	Factual Premise	An empirical proposition used to support ethical reasoning.
ERC03	**ERC**	Normative Premise	An explicit moral principle, value, duty, or norm supporting the reasoning.
ERC04	**ERC**	Conclusion	An explicit ethical conclusion derived from the reasoning.
ERC05	**ERC**	Counterargument	Recognition of an alternative position or objection.
ERC06	**ERC**	Moral Conflict	Explicit recognition of competing moral considerations.
ERC07	**ERC**	Deliberative Weighing	Explicit comparison or balancing of competing considerations before reaching a conclusion.
ERC08	**ERC**	Resolution	Proposal of a justified course of action.
MSC01	**MSC**	Welfare / Well-being	Explicit consideration of physical or psychological well-being, suffering, harm, or benefit.
MSC02	**MSC**	Rights	Explicit recognition of rights, entitlements, or moral claims.
MSC03	**MSC**	Justice / Fairness	Explicit concern for fairness, equality, equity, or distributive justice.
MSC04	**MSC**	Autonomy	Explicit recognition of agency, freedom, self-determination, or informed choice.
MSC05	**MSC**	Responsibility / Duty	Explicit appeal to obligations, duties, accountability, or professional responsibilities.
MSC06	**MSC**	Vulnerability	Explicit recognition of fragility, dependency, marginalization, or the need for protection.
MSC07	**MSC**	Relationships / Care	Explicit emphasis on interpersonal relationships, empathy, care, trust, or relational responsibilities.
MSC08	**MSC**	Intrinsic Value	Explicit recognition that an individual, organism, or entity possesses value independent of its utility.
MSC09	**MSC**	Capabilities / Flourishing	Explicit concern for the ability to develop, function, or flourish.
MSC10	**MSC**	Social or Cultural Context	Explicit recognition that social, cultural, historical, political, or institutional conditions are morally relevant.
SJ01	**SJ**	Empirical / Scientific Evidence	Explicit reliance on empirical data, research findings, or scientific knowledge.

ID	Dimension	Variable	Operational Definition
SJ02	**SJ**	Professional / Clinical Expertise	Explicit appeal to professional training, clinical experience, or disciplinary expertise.
SJ03	**SJ**	Philosophical / Theoretical Concepts	Explicit use of philosophical theories, ethical frameworks, or theoretical concepts.
SJ04	**SJ**	Legal / Regulatory Norms	Explicit appeal to laws, regulations, or formal institutional norms.
SJ05	**SJ**	Personal Experience	Explicit reference to the participant's own lived experience.
SJ06	**SJ**	Cultural / Traditional Norms	Explicit appeal to cultural practices, customs, or tradition.
SJ07	**SJ**	Religious / Spiritual Beliefs	Explicit appeal to religious doctrine, spiritual belief, or faith-based reasoning.
SJ08	**SJ**	Analogical Reasoning	Explicit use of an analogy or comparison to justify the judgment.
SJ09	**SJ**	Other Explicit Source	Any other explicitly identifiable source of justification.

Appendix B. Decision Trees

This appendix provides decision trees for six recurring sources of coding ambiguity. Coders should consult these before logging a case as unresolved (Rule 7, Section 9).

Decision Tree 1. ERC01 (Ethical Position) vs. ERC04 (Conclusion)

- 1. Does the response state what ought or ought not to be done, or evaluate an action, institution, or practice in moral terms? If no, do not code ERC01. If yes, continue.
- 2. Is this statement presented as the outcome of preceding reasoning (i.e., following premises or explicit weighing)? If no, code ERC01 only. If yes, code both ERC01 and ERC04 — a conclusion may coincide with an ethical position (see the Notes column for ERC04 in Section 8).

Decision Tree 2. MSC01 (Welfare) vs. MSC09 (Capabilities / Flourishing)

- 1. Does the response reference pain, suffering, harm, or benefit? If yes, code MSC01.
- 2. Does the response separately reference the ability to develop, function, or flourish beyond the avoidance of harm (e.g., living according to one's nature, capabilities)? If yes, code MSC09 in addition to MSC01, where applicable. If no, do not code MSC09.

Decision Tree 3. SJ01 (Empirical Evidence) vs. SJ08 (Analogical Reasoning)

- 1. Does the response cite data, research findings, or factual/scientific information? If yes, code SJ01.
- 2. Does the response compare the dilemma to a different case or situation to support the judgment? If yes, code SJ08.
- 3. Both SJ01 and SJ08 may be coded 1 in the same response if both elements are present independently of each other.

Decision Tree 4. ERC02 (Factual Premise) vs. ERC03 (Normative Premise)

- 1. Does the statement describe what is empirically the case (a fact, observation, or finding) used to support the reasoning? If yes, code ERC02.
- 2. Does the statement instead prescribe what ought or ought not to be the case, or state a value, duty, or principle? If yes, code ERC03.
- 3. A single response may contain both an explicit factual claim and an explicit normative claim; code both ERC02 and ERC03 when each is separately and explicitly present. Do not code ERC03 merely because a factual claim is used in service of an ethical position — the normative content itself must be explicit.

Decision Tree 5. MSC02 (Rights) vs. MSC04 (Autonomy)

- 1. Is the consideration framed as an entitlement or claim the participant holds against others (e.g., a right to something)? If yes, code MSC02.
- 2. Is the consideration instead framed in terms of agency, self-determination, or the capacity to choose, without an explicit rights claim? If yes, code MSC04.
- 3. When autonomy is explicitly framed as a right (e.g., ‘the right to make one's own decisions’), code both MSC02 and MSC04.

Decision Tree 6. MSC05 (Responsibility / Duty) vs. SJ04 (Legal / Regulatory Norms)

- 1. Does the response recognize an obligation, duty, or accountability as morally relevant to the dilemma, regardless of its source? If yes, code MSC05.
- 2. Does the response explicitly cite a law, regulation, or formal institutional norm as the grounds for the ethical judgment? If yes, code SJ04.
- 3. MSC05 and SJ04 are not mutually exclusive: a response that recognizes a duty (MSC05) and explicitly grounds that duty in a legal or regulatory norm (SJ04) should be coded 1 on both, consistent with Rule 2 (Section 9) — MSC codes what is recognized as relevant, SJ codes the source invoked to justify it.

Appendix C. Annotated Responses

The following responses are constructed for training purposes and are not drawn from a specific dataset. They supplement the worked example in Section 11.

Annotated Response 1 — Dilemma: resource allocation in public health

"It is not fair that people with more money get faster access to treatment. Everyone should have an equal chance regardless of their ability to pay, because health is a basic need, not a privilege. I know this is complicated to implement, but a lottery system for scarce treatments would at least be more just than a system based on who can pay first."

- ERC01 (Ethical Position) = 1 — explicit normative stance against ability-to-pay allocation.
- ERC03 (Normative Premise) = 1 — "health is a basic need, not a privilege."
- ERC08 (Resolution) = 1 — proposes a lottery system as a practical course of action.
- MSC03 (Justice / Fairness) = 1 — explicit and central concern for fairness in allocation.

Annotated Response 2 — Dilemma: confidentiality in professional practice

"In my training we were always taught that confidentiality is almost absolute, but I think there has to be an exception when someone else is at serious risk of harm. It is a hard balance: on one hand you owe loyalty and trust to the person you are helping, on the other hand you have a responsibility to prevent harm to a third party. I would only break confidentiality if the risk were immediate and severe."

- ERC06 (Moral Conflict) = 1 — confidentiality/trust explicitly framed as in tension with preventing harm.
- ERC07 (Deliberative Weighing) = 1 — "it is a hard balance", weighing both sides explicitly.
- ERC08 (Resolution) = 1 — conditional course of action specified (immediate and severe risk).
- MSC05 (Responsibility / Duty) = 1 — "responsibility to prevent harm to a third party."
- MSC07 (Relationships / Care) = 1 — "loyalty and trust to the person you are helping."
- SJ02 (Professional / Clinical Expertise) = 1 — "in my training we were always taught…"

Appendix D. Example Dataset

The table below extends the simplified example in Section 5 to illustrate how a coded dataset looks in practice, using a subset of variables. A full ERM dataset would include all 27 ERC/MSC/SJ columns following the same binary (0/1) structure.

response_id	participant_id	timepoint	dilemma_id	ERC01	ERC07	MSC01	MSC03	SJ01	SJ08
R001	P001	Pre	D01	1	0	1	0	0	0
R002	P001	Post	D01	1	1	1	1	0	1
R003	P002	Pre	D01	0	0	0	0	0	0
R004	P002	Post	D01	1	1	1	1	1	0
R005	P003	Pre	D02	1	0	0	1	0	0
R006	P003	Post	D02	1	1	1	1	1	1

Appendix E. R Scripts

The script below implements the R Workflow summarized in Section 14, using the tidyverse and irr packages.

```
library(tidyverse)



# 1. Import coded data

erm_data <- read_csv("data/coded/erm_coded_data.csv")



# 2. Define variable set and validate binary coding

erm_vars <- c(paste0("ERC0", 1:8),

              paste0("MSC0", 1:9), "MSC10",

              paste0("SJ0", 1:9))



erm_data %>%

  summarise(across(all_of(erm_vars), ~ all(. %in% c(0, 1)))) %>%

  pivot_longer(everything(), names_to = "variable", values_to = "is_binary")



# 3. Compute dimension indices

erm_data <- erm_data %>%
```

```
  mutate(
    ERC_index = rowSums(across(starts_with("ERC"))),
    MSC_index = rowSums(across(starts_with("MSC"))),
    SJ_index  = rowSums(across(starts_with("SJ")))
  )

# 4. Variable-level frequencies by timepoint, with 95% Wilson confidence intervals
library(binom)
erm_freq <- map_dfr(erm_vars, function(v) {
  x <- erm_data[[v]]; x <- x[!is.na(x)]
  ci <- binom.confint(sum(x), length(x), conf.level = 0.95, methods = "wilson")
  tibble(variable = v, proportion_present = sum(x) / length(x),
         ci_lower = ci$lower, ci_upper = ci$upper)
})

# 5. Inter-rater reliability (pilot double-coded subsample)
library(irr)
# Two coders, nominal binary data: unweighted Cohen's kappa (Cohen, 1960)
kappa2(erm_data %>% select(coder1_ERC01, coder2_ERC01), weight = "unweighted")
```

```
# > 2 coders, or missing ratings: Krippendorff's alpha (Krippendorff, 2018)

kripp.alpha(t(erm_data %>% select(coder1_ERC01, coder2_ERC01, coder3_ERC01)),

            method = "nominal")



# 6. Longitudinal comparisons

# Binary variable, two timepoints, same participants: McNemar's test (McNemar, 1947)

mcnemar.test(table(erm_data$ERC01_pre, erm_data$ERC01_post))

# Dimension index, two timepoints: Wilcoxon signed-rank test (Wilcoxon, 1945)

wilcox.test(ERC_index ~ timepoint, data = erm_data, paired = TRUE)

# Pairwise co-occurrence tests: control the false discovery rate (Benjamini &
Hochberg, 1995)

p_values <- combn(erm_vars, 2, \(v)

  fisher.test(table(erm_data[[v[1]]], erm_data[[v[2]]]))$p.value)

p.adjust(p_values, method = "BH")



# 7. Visualization: dimension index by timepoint

library(ggplot2)

erm_data %>%

  group_by(timepoint) %>%

  summarise(mean_ERC = mean(ERC_index)) %>%
```

```
  ggplot(aes(x = timepoint, y = mean_ERC, group = 1)) +

  geom_line() + geom_point() +

  labs(title = "ERC Index by Timepoint", y = "Mean ERC Index", x = "Timepoint")



# 8. Export summary tables and figures for reporting

# (see scripts/erm_analysis.R in the companion ERM_project for a complete,

#  exportable implementation of steps 1-8)
```

For a complete, production-ready implementation of this pipeline — including input validation, exported tables and figures, and automated reliability reporting across all 27 variables — see scripts/erm_analysis.R in the versioned GitHub release described under Companion Materials, below.

Appendix F. Changelog

Version	Date	Changes
1.0	August 2026	Initial complete public release. Added the Sources of Justification variable table (SJ01–SJ09), which was referenced throughout the manual but previously unspecified. Added Sections 9–16 (Coding Rules, Quality Control, Worked Examples, Variable Dictionary, Data Outputs, R Workflow, File Structure, Version History). Added Appendices A–F (Complete Codebook, Decision Trees, Annotated Responses, Example Dataset, R Scripts, Changelog). Added a References section and title-page metadata for open deposit. Harmonized dimension labels (e.g., ‘Normative Considerations’ relabeled ‘Morally Salient Considerations (MSC)’ for consistency with Sections 3 and 7).
1.1	August 2026	Added Alma Delia Torres-Rivera as co-author (title page and suggested citation). Strengthened methodological and psychometric rigor: specified minimum pilot sample size and its statistical rationale (Bujang & Baharum, 2017; O’Connor & Joffe, 2020); specified the exact kappa/alpha estimators, R implementations, and the conventional basis of the $\kappa/\alpha \geq .70$ threshold (Cohen, 1960; Krippendorff, 2018; Hayes & Krippendorff, 2007; Landis & Koch, 1977); added a re-piloting trigger to ongoing monitoring; added a Validity Considerations subsection to Section 10. Added conceptual precision: cross-referenced boundary notes in Section 8 and three new decision trees in Appendix B (ERC02 vs. ERC03; MSC02 vs. MSC04; MSC05 vs. SJ04). Added Rule 9 (Section 9) specifying the review and promotion procedure for SJ09. Added statistical specification to Section 13 (confidence intervals, McNemar's test, Wilcoxon signed-rank test, mixed-effects models for clustering, false discovery rate correction for pairwise co-occurrence tests) and aligned Section 14 (R Workflow) and Appendix E accordingly. Expanded the References section with nine new citations supporting the above additions. Added a Companion Materials section (before References) pointing to the versioned GitHub repository and its v1.1 release, and replaced the informal local-path pointer in Appendix E with a reference to it. Finalized the license statement on the title page (CC BY 4.0, approved by both authors). Removed an inconsistent SJ03 annotation in Appendix C, Annotated Response 1 (the response does not explicitly name a theory, so SJ03 should not have been coded 1). Completed the Appendix E code for variable-level frequencies so it actually computes Wilson confidence intervals, matching its own description. Added a Validation Status subsection to Section 10 stating that this version has not yet undergone expert-panel, pilot, or inter-rater validation.